

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Technical Presentation

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Duration: 60 Minutes

Total Marks: 25

Code of Conduct

I ____ G.Jaahnavi -192525277____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____ G.Jaahnavi ____

Criteria	Technical Content (1.5)	Structure and Flow (1)	Use of Visuals (0.75)	Communication Skills (1)	Professionalism (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Assessment Tasks

Technical Presentation Topics

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

1. Describing the Role of Data Centers in Cloud Computing - Interpret how data centers support cloud services.
2. Explaining the Role of Web Applications in Cloud Computing - Interpret how cloud-based applications function over the internet.
3. Describing the Role of Web Browsers in Accessing Cloud Services - Interpret how browsers act as client interfaces.
4. Understanding Cloud Storage Mechanisms - Explain how data is stored, managed, and accessed in the cloud.
5. Describing the Role of Networking in Cloud Computing - Interpret how network infrastructure supports cloud services.

Minimum slide flow:

- Title and objective
- Core technical explanation
- Architecture or process diagram
- Real-world use case
- Conclusion and key takeaway

Technical Presentation Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content	30%	Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Content is inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism



SIMATS
ENGINEERING



SIMATS
Sreea Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CloudGuard X: Autonomous Cloud Security & Self-Healing Platform

CSA-1521 CLOUD COMPUTING

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Introduction

- Cloud computing stores data and applications on the internet.
- Many organizations use cloud services for flexibility and cost savings.
- However, cloud environments face cyber threats like hacking, malware, insider attacks, and data breaches.
- Traditional security systems detect threats only after they occur.
- CloudGuard X is an AI-based security platform that continuously monitors cloud activities, predicts threats, and automatically protects cloud resources.



Problem Statement

- Increasing number of cyber attacks in cloud environments.
- Delay in detecting security threats.
- Manual monitoring is time-consuming.
- Difficult to identify abnormal user behavior.
- Slow incident response may lead to data loss and service interruption.



LITERATURE REVIEW



Author(s)	Paper Title	Year	Methods Used	Inference
Samuel O. Olabanji, Yewande A. Marquis, Chinasa S. Adigwe, et al.	AI-Driven Cloud Security: Examining the Impact of User Behavior Analysis on Threat Detection	2024	Artificial Intelligence (AI), User Behaviour Analytics (UBA), Machine Learning, Threat Prediction, Pattern Recognition (Asian Journal of Computer Science)	Improved threat detection by analyzing user behavior and identifying suspicious patterns using AI and machine learning.
Yehia Ibrahim Alzoubi, Alok Mishra, Ahmet Ercan Topcu	Research Trends in Deep Learning and Machine Learning for Cloud Computing Security	2024	Machine Learning (ML), Deep Learning (DL), Anomaly Detection, Security Automation, Behavioral Analytics (Springer)	Showed that ML and DL techniques can effectively detect anomalies and strengthen automated cloud security.
Luna Alexandra	Self-Healing Cloud Security Systems: Integrating AI with Zero Trust for Real-Time Threat Mitigation	2023	Artificial Intelligence, Zero Trust Architecture, Behavioral Analytics, Automated Remediation, Self-Healing Mechanism (ResearchGate)	Demonstrated real-time threat mitigation through AI, Zero Trust, automated remediation, and self-healing mechanisms.
Haili Wang, Jingda Guo, Xu Ma, Song Fu, Qing Yang, Yunzhong Xu	Online Self-Evolving Anomaly Detection in Cloud Computing Environments	2021	Self-Evolving Anomaly Detection (SEAD), Online Learning, Runtime Monitoring, Machine Learning (arXiv)	Improved anomaly detection by continuously learning from changing cloud environments and adapting to new threats.
Sabbir M. Saleh, Ibrahim Mohammed Sayem, Nazim Madhavji, John Steinbacher	Advancing Software Security and Reliability in Cloud Platforms through AI-based Anomaly Detection	2024	CNN, LSTM, AI-based Anomaly Detection, Network Traffic Analysis, CI/CD Security Monitoring (arXiv)	Enhanced cloud security and reliability by using AI-based anomaly detection and continuous security monitoring.



Objectives



- Monitor cloud activities continuously.
- Collect and analyze cloud logs.
- Detect suspicious user behavior.
- Predict cyber threats using AI.
- Identify anomalies in cloud systems.
- Automatically prevent attacks.
- Generate alerts and security reports.

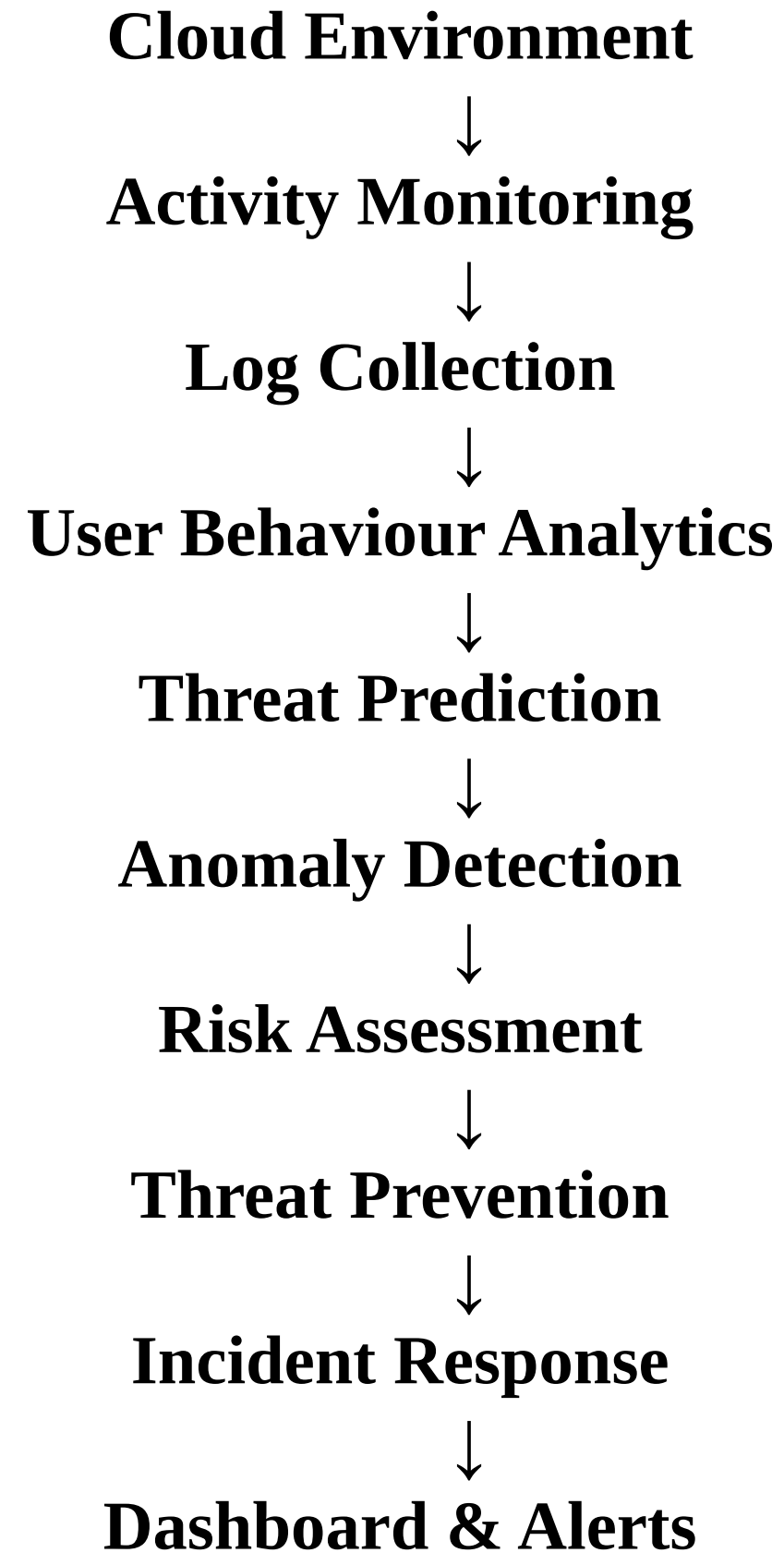


METHODOLOGY

- The cloud platform continuously monitors all user activities and system events.
- User login details, logout details, file uploads, downloads, and network activities are captured.
- System logs, application logs, and security logs are collected and stored in a secure database.
- The collected log data is organized and prepared for further analysis.
- User behavior is analyzed using historical activity data to understand normal usage patterns.
- Current user activities are compared with previous behavior to identify unusual actions.



System Architecture





Module 1: Cloud Activity Monitoring & Log Collection

Purpose

The main purpose of this module is to continuously monitor all activities happening in the cloud environment and store them as log files. These logs provide the basic information required to identify security threats in later modules.

Input

- User login requests
- File upload and download activities
- Server access information
- Network traffic
- API requests
- System events

Output

- User activity logs
- System logs
- Network logs
- Event logs

Process

1. Monitor every user activity in the cloud.
2. Record login and logout information.
3. Track file access and modifications.
4. Monitor server and network events.
5. Store all collected information in a secure log database.
6. Organize the logs with date and time for future analysis.



Module 2: User Behaviour Analytics & Threat Prediction

Purpose

This module analyzes the activity logs received from Module 1 to understand normal user behavior and identify suspicious activities.

Input

- User activity logs
- Login history
- Device information
- Location details
- Previous user behavior

Output

- User behavior report
- Threat prediction report
- Suspicious user list
- Threat score

Process

1. Read the activity logs from Module 1.
2. Build a profile of normal user behavior.
3. Compare current activities with previous activities.
4. Identify unusual login times, locations, or devices.
5. Predict possible cyber threats using AI or machine learning.
6. Generate a threat score for each user.



Module 3: Anomaly Detection & Risk Assessment

Purpose

This module detects abnormal activities and calculates the level of security risk.

Input

- Threat prediction report
- User behavior report
- Activity logs

Output

- Anomaly report
- Risk score
- High-risk alerts
- Risk assessment report

Process

1. Compare current activities with normal behavior.
2. Detect unusual patterns such as repeated login failures or large data downloads.
3. Identify unauthorized access attempts.
4. Calculate the risk score based on the detected anomalies.
5. Classify the risk as Low, Medium, or High.
6. Prepare a risk assessment report.



Module 4: Automated Threat Prevention & Incident Response



Purpose

This module automatically responds to detected threats and protects the cloud environment without waiting for manual action.

Input

- Risk assessment report
- High-risk alerts
- Anomaly report

Output

- Threat prevention report
- Incident response report
- Recovery status
- Action log

Process

1. Receive the high-risk alert from Module 3.
2. Block suspicious user accounts.
3. Terminate unauthorized sessions.
4. Isolate affected cloud resources.
5. Send OTP or multi-factor authentication requests if needed.
6. Record all actions taken during the incident.
7. Restore affected services through self-healing mechanisms.



Module 5: Cloud Security Dashboard, Reporting & Alert Management

Purpose

This module provides a centralized dashboard for monitoring cloud security and generating reports and alerts.

Input

- Incident response report
- Threat prevention report
- Recovery status
- Security logs

Output

- Real-time security dashboard
- Security reports
- Alert notifications
- Cloud security status

Process

- Collect security information from previous modules.
- Display real-time cloud security status.
- Show detected threats and blocked attacks.
- Generate daily, weekly, and monthly security reports.
- Send alerts through email, SMS, or mobile notifications.
- Maintain a history of all security incidents for future analysis.



Expected Results



- Detect cyber threats at an early stage.
- Prevent attacks automatically before major damage occurs.
- Minimize downtime using self-healing mechanisms.
- Keep cloud data and services secure.
- Help administrators monitor the cloud efficiently through a centralized dashboard.



Advantages



- Improves cloud security.
- Detects attacks quickly.
- Reduces manual monitoring.
- Minimizes data breaches.
- Fast incident response.
- Increases system availability.
- Supports business continuity.
- Cost-effective security management.



Applications



- Banking and Finance
- Healthcare
- E-commerce
- Educational Institutions
- Government Organizations
- IT Companies
- Cloud Service Providers
- Enterprise Data Centers



Conclusion

- **Continuous Cloud Monitoring:** Continuously monitors cloud infrastructure, applications, users, and network activities to identify security risks in real time.
- **Threat Prediction & Anomaly Detection:** Uses intelligent analysis to predict potential cyber threats and detect unusual activities that may indicate security attacks.
- **Automatic Threat Response:** Automatically identifies and responds to attacks by blocking malicious activities and isolating affected cloud resources.
- **Self-Healing Capability:** Automatically recovers affected systems and restores cloud services after an attack, reducing downtime and manual recovery efforts.
- **Reliable & Secure Cloud Services:** Improves overall cloud security, minimizes service interruptions, and ensures reliable operation with minimal human intervention.



References

- S. V. Chennupati and S. Banerjee, “Machine Learning Approaches for Threat Detection in Cloud Computing,” in *Proc. 2025 IEEE 5th Int. Conf. on ICT in Business Industry & Government (ICTBIG)*, 2025, doi: 10.1109/ICTBIG68706.2025.11323906.
- “Protecting Cloud Computing Environments from Cyber Threats with AI-Powered Machine Learning Systems,” in *Proc. 2025 Global Conf. in Emerging Technology (GINOTECH)*, Pune, India, May 9–11, 2025, doi: 10.1109/GINOTECH63460.2025.11076781.
- “Intrusion Detection System in Clouds Using Machine Learning,” in *Proc. 2025 5th Int. Conf. on Intelligent Technologies (CONIT)*, 2025, doi: 10.1109/CONIT65521.2025.11167228.
- S. M. Shaffi, S. Vengathattil, J. N. Sidhick, and R. Vijayan, “AI-Driven Security in Cloud Computing: Enhancing Threat Detection, Automated Response, and Cyber Resilience,” 2025, arXiv:2505.03945.
- C. Ji and H. Luo, “Cloud-Based AI Systems: Leveraging Large Language Models for Intelligent Fault Detection and Autonomous Self-Healing,” 2025, arXiv:2505.11743.



**THANK
YOU**